

<https://doi.org/10.1038/s43247-025-02855-4>

Interlacing ruptures of the 2024 Wushi earthquake (Chinese Tian Shan) controlled by structural inheritance



Xinzhong Yin^{1,2}, Tao Li^{1,2}✉, Yingfeng Zhang^{1,2}✉, Zhigang Peng³, Luca Dal Zilio^{4,5}, Zhuxin Chen⁶, Huawei Cui⁷, Li Qian^{1,2}, Xiao Sun^{1,2}, Renqi Lu^{1,2}, Biao Guo^{1,2}, Wenxin Yang^{1,2}, Jiuhui Chen^{1,2} & Jie Chen^{1,2}

The 2024 M_w 7.0 Wushi earthquake is the largest earthquake within the seismically active Tian Shan during the past three decades. Through integrated analyses of seismic, geodetic, and geologic data, here we investigate the fault structure corresponding to the Wushi mainshock and subsequent aftershocks. Our work reveals (i) a moderately-dipping, oblique-reverse main-fault rupture and (ii) numerous steeply to vertically-dipping subfault ruptures that align obliquely in cross-section to the main fault. The subfaults made clear aftershock concentrations while the main fault did not, and a shallowly-buried subfault generated clear surface rupture despite the main fault being totally blind. This fault network, misoriented with the prevailing background stress field, likely forms through a reactivation of inherited planes of weakness. These results demonstrate the significant control of structural inheritance on fault geometric architecture and highlight the complexity of seismic activity and rupture behavior of the yielding fault network.

Earthquakes represent rapid slip along faults that contain a wide range of structural geometries and complexities¹. While mature faults and associated large earthquake ruptures generally exhibit relatively simple geometries with minor splays (e.g., the 2002 M_w [moment magnitude] 7.8 Alaska² and the 2025 M_w 7.7 Myanmar earthquakes³), there is growing evidence arguing for much more complex earthquake ruptures with numerous oblique or even orthogonal subsidiary faults, such as conjugate faults (e.g., the 2019 Ridgecrest earthquake sequence⁴), antithetic fault-tip splays⁵, as well as hanging-wall accommodation (e.g., the 1983 M_w 6.5 Coalinga earthquake sequence⁶) or transverse (e.g., the 2020 M_w 6.0 Jiashi earthquake sequence⁷) faults specific for dip-slip events. These subsidiary faults can interplay with the main fault to generate a complicated multi-fault mainshock rupture^{8,9} and/or fore-main-aftershock sequence^{4,7,10–15}. Furthermore, they can generate aftershock clusters that significantly deviate from the main fault^{4,16–20}; these subfault slip events, however, mostly occur outside the main coseismic slip area^{4,16,20}, i.e., resulting in a general anti-correlation distribution pattern of the coseismic slip and the aftershocks^{21–23}. The interlacing fault system can develop in a variety of tectonic settings. A thorough understanding of its structural appearance, origin, and evolution during large earthquake sequences is crucial for deciphering the relationship between

fault structures and fault slip behaviors, and for better assessing potential seismic hazards.

On 22 January (UTC time), 2024, a M_w 7.0 earthquake struck the Wushi basin along the southern margin of Chinese Tian Shan (Fig. 1a). Earthquake parameters from different agencies (Supplementary Data 4) indicate an epicenter of (41.256°N, 78.654°E), a focal depth of 13–22 km, and a moderate-angle oblique-reverse faulting mechanism (strike/dip/rake of the NW-dipping nodal plane: 233–235°/45–49°/41–44°). This earthquake was followed by numerous strong aftershocks; by 7 March 2024, more than 150 aftershocks of $M \geq 4.0$ were recorded, most of which feature similar focal mechanisms to the mainshock (Fig. 2a). Field survey after the mainshock revealed no clear mainshock-related surface faulting rupture²⁴, consistent with relatively continuous fringes of the InSAR (Interferometric Synthetic Aperture Radar) observations^{25,26}. However, a clear surface rupture zone (with a length of >2 km and a maximum vertical offset of >1 m) was produced by the M_w 5.8 aftershock that occurred 7 days after the mainshock (January 29) on a SE-dipping reverse fault^{24,27} (strike/dip/rake of the SE-dipping nodal plane: 40–76°/32–59°/40–116°). The Wushi earthquake is the largest earthquake since the 1992 M_w 7.3 Suusamy, Kyrgyzstan event, and it is also one of the few large earthquakes that were recorded by a

¹Xinjiang Pamir Intracontinental Subduction National Observation and Research Station, Institute of Geology, China Earthquake Administration, Beijing, China.

²Lhasa National Geophysical Observation and Research Station, Institute of Geology, China Earthquake Administration, Beijing, China. ³School of Earth and Atmospheric Sciences, Georgia Institute of Technology, Atlanta, GA, USA. ⁴Earth Observatory of Singapore, Nanyang Technological University, Singapore, Singapore.

⁵Asia School of the Environment, Nanyang Technological University, Singapore, Singapore. ⁶Research Institute of Petroleum Exploration and Development, PetroChina, Beijing, China. ⁷Shandong Earthquake Agency, Jinan, China. ✉e-mail: lita.410@163.com; jingqing129@gmail.com

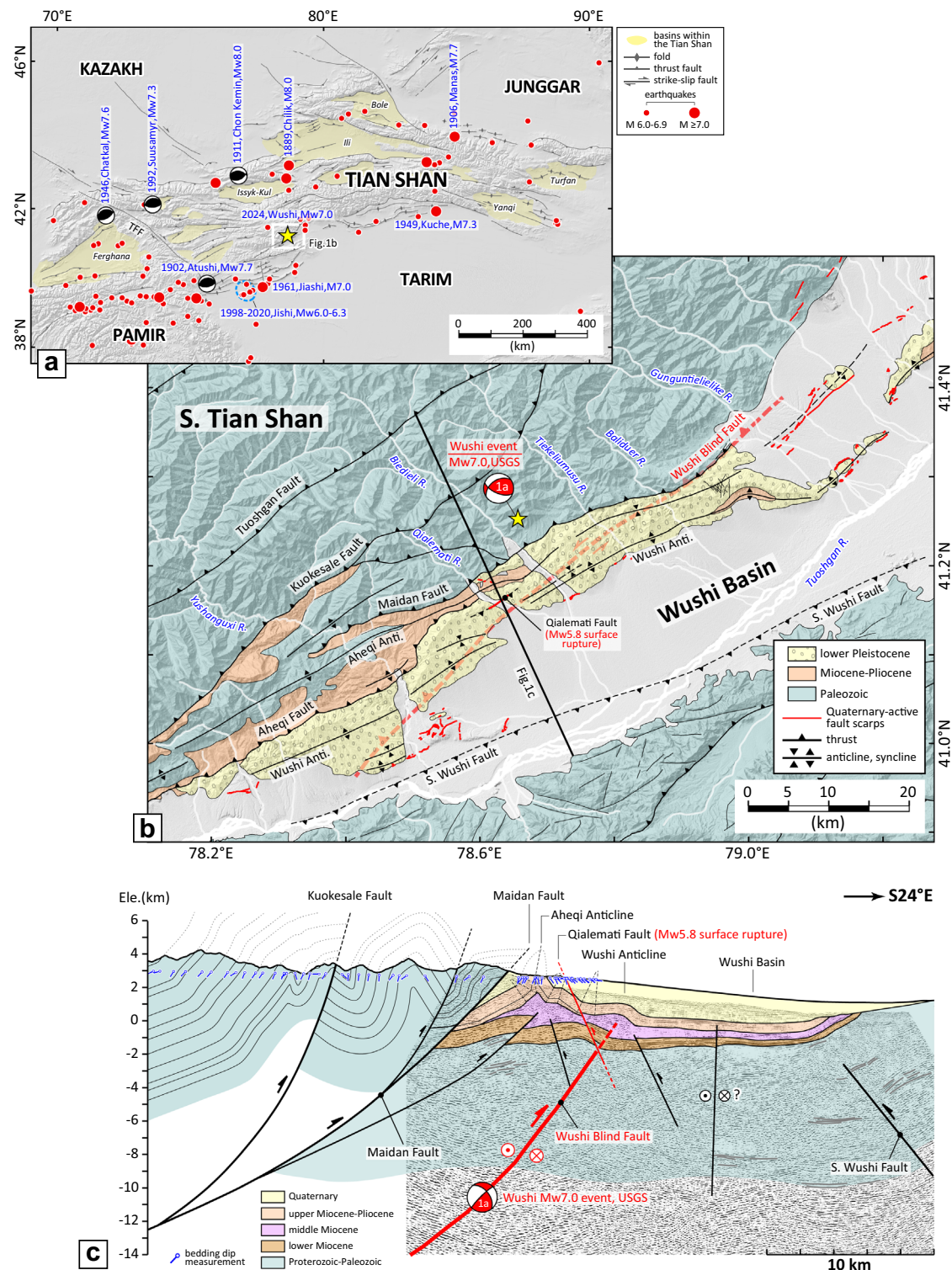


Fig. 1 | Seismotectonic background of the 2024 Wushi earthquake. a Topography, active structures (modified from Hu et al.⁸³), and major earthquakes (from 1885 to August 2024) within the Tian Shan and adjacent regions. Earthquake epicenters and focal mechanisms from the U. S. Geological Survey and Kulikova³⁹. TFF: Talas-Fergana Fault. **b** Geologic map of the Wushi region (location shown in Fig. 1a) based on a 1:200,000 geologic map (made in 1957–1961 by Xinjiang Bureau of Geology and Mineral Resources) and our field mapping. The trace of the Qialemati Fault based on InSAR observations (Fig. 2b) and the M_w 5.8 (on January 29) surface rupture from Zhang et al.²⁴. **c** Geologic section (location shown in Fig. 1b) across the southern Tian

Shan and the Wushi basin, interpreted from a petroleum seismic-reflection profile (base map; see Methods & Supplementary Fig. 14) and our field mapping data, and its correlation with the InSAR-inverted mainshock and M_w 5.8 aftershock rupture geometries (highlighted using red lines). The topographic line is extracted from the Shuttle Radar Topography Mission DEM. Depth conversion of the seismic-reflection profile uses a uniform velocity of 5000 m/s, such that the Cenozoic sedimentary sequence and hosting structures may be vertically exaggerated by 1.4 due to their lower velocity (3500 m/s).

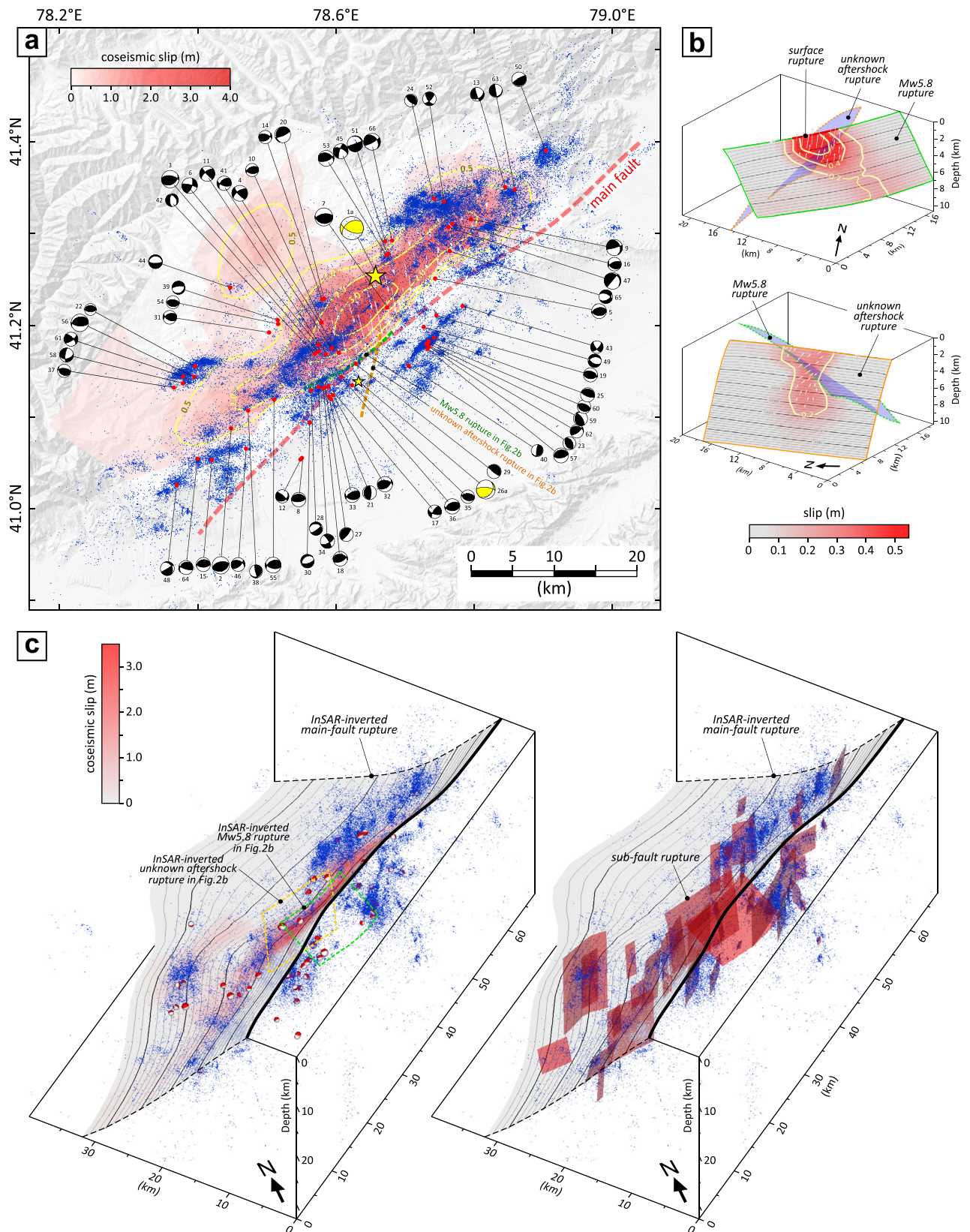


Fig. 2 | Map view and 3D view of Wushi seismicity and causative fault geometry. **a** Map view of earthquake epicenters, focal mechanisms (listed in Supplementary Data 3–4), and surface traces of InSAR-inverted sub-fault ruptures in Fig. 2b superimposed on InSAR-inverted mainshock slip distribution.

b Geometry and slip distribution of two shallow sub-fault ruptures (occurred between January 26 and February 5, 2024) based on InSAR inversions. **c** 3D view showing relations of earthquake hypocenters, subsidiary fault geometries, and the main fault geometry.

high-quality observation system (including a dense seismic network and SAR satellites), within the Tian Shan region.

At the time of the mainshock, all seismic stations were located at more than 50 km from the epicenter. Immediately following the mainshock, a very dense local network of 76 seismic stations was installed to improve near-field and azimuthal coverage of the seismicity region (see Methods & Supplementary Fig. 1). In this study, we take advantage of the recorded seismic data to develop a high-resolution aftershock catalog (59,574 events with uncertainties of 80 m in epicenter and of 91 m in focal depth) with machine-learning-based phase picking/association and waveform-based relocation techniques (see Methods). This dataset, combining with (i) InSAR data inversions, (ii) well-constrained earthquake focal mechanisms, and (iii) detailed surface mapping data and petroleum seismic-reflection profiles, delineates a complicated fault network ruptured during the Wushi earthquake sequence. We then analyze the fault structural origin, as well as its correlation with aftershock activities, particularly the M_w 5.8 event. Our study demonstrates that the Wushi earthquake fault contains some unique geometric features and earthquake slip behaviors. Additionally, it also offers critical insights into the seismogenic characteristics of the Tian Shan region.

Seismotectonic background

The Tian Shan mountain, extending over 2500 km between the Tarim basin to its south and the Kazakh platform-Junggar basin to its north (Fig. 1a), has undergone significant Cenozoic shortening and uplift due to the India-Asia collision²⁸. Currently, it maintains a crustal shortening rate of 5–20 mm/a^{29–32}, which is mainly accommodated by numerous moderately-to-steeply dipping, basement-rooted reverse and oblique-reverse faults in the hinterland^{33–35}, and an array of mountain-parallel thin-skinned thrust systems in the foreland^{36–38}. The Tian Shan region had suffered a remarkable sequence of large earthquakes from 1885 to 1946, during which totally five earthquakes of $M \geq 7.5$ occurred^{39,40} (Fig. 1a). However, it had become relatively quiescent since 1947, and only several M 7.0–7.4 events (e.g., the 1961 M 7.0 Jiashi and the 1992 M_w 7.3 Susamyr events) occurred prior to the 2024 Wushi earthquake.

The Wushi basin, hosting the 2024 Wushi event, lies along the Tian Shan southern margin (Fig. 1a). The basin opens eastward toward the Tarim basin, but tapers gradually to the west (Fig. 1b); on the north and south margins, it is bounded by opposite-vergence reverse faults (the Maidan Fault and the Southern Wushi Fault, respectively) and fault-uplifted Paleozoic exposures. Sedimentary fills within the basin include a 2–4-km-thick, coarsening-upward Cenozoic terrestrial sequence that is underlain along an angular unconformity by strongly-deformed Paleozoic limestone, dolomite and mudstone and Proterozoic metamorphic deposits^{41,42} (Fig. 1c). The middle-Miocene unit, comprising salt, gypsum and mudstone, is notably weak and mobile, serving as a regional detachment horizon that thickens at anticlinal cores and tapers off toward the limbs. The subsalt structures are characterized by multiple high-angle NE- or SW-dipping reverse or strike-slip faults, whose dip-slip displacements are estimated to be several hundreds to thousands of meters (Fig. 1c). In contrast, the suprasalt structures consist of an array of NE-trending, broad to tight detachment or fault-propagation folds (e.g., the Aheqi and Wushi anticlines in Fig. 1b, c). These shallow structures generate numerous offset geomorphic features at the surface, indicating ongoing deformation throughout the Quaternary⁴³ (Fig. 1b).

For the faults within the Wushi basin, several previous studies^{43,44} interpreted that all of them cut through the entire Proterozoic-Cenozoic sequence and expose at the surface, and these faults were defined as the branches of Maidan Fault, a >800-km-long fault along the Tian Shan southern margin⁴⁵. However, our interpretation of the seismic-reflection profile clearly shows that the subsalt and suprasalt structures have both a decoupling deformation pattern and a limited deformation magnitude, significantly different from the Maidan Fault. Therefore, in this study, we separate these structures (also including the Wushi earthquake fault, see below) from the Maidan Fault that refers to Zheng et al.⁴⁶ and name them independently.

Results

Based on teleseismic data inversions from the U. S. Geological Survey and our InSAR data analyses (Fig. 2 & 3b, c; Methods), the Wushi mainshock (M_w 7.0) rupture initiated near Biediel valley, and then propagated 29 km southwestward and 22 km northeastward along a NW-dipping fault plane. The SW fault segment maintains a relatively constant dip of 50°; in comparison, the NE segment exhibits a curved geometry, dipping 30–35° below 17 km depth and steepening to 55–60° above 14 km. The resolved coseismic slip, with approximately equal dip-slip and sinistral-slip components, is up to 3–4 m near the mainshock epicenter and decreases gradually to the two fault tips. In the downdip direction, the slip is concentrated at depths of shallower than 20 km but deeper than 3–10 km (Figs. 2c & 3b), consistent with the interpretation of a blind fault rupture^{24–26}.

Our precisely-located aftershocks are distributed across an 80 km by 25 km NE-trending area (Fig. 2a). They show no clear concentration along the InSAR-determined mainshock fault (Figs. 2c & 3), but rather express as a suite of planar or quasi-planar clusters, dipping steeply (69–90°) and aligning obliquely with the main fault in the cross-section view. These seismic clusters indicate the presence of numerous subsidiary fault ruptures across various length scales and orientations following the mainshock (Fig. 3; Methods & Supplementary Fig. 16). At the SW corner of the mainshock rupture (Fig. 3c profile 1), the clusters are mostly located at 2–15 km depth in the main-fault's hanging wall. They are roughly NE-NEE oriented, and from which 9 faults of 0.7–8 km long are interpreted. At the SW segment of the mainshock rupture (Fig. 3c profiles 2–3), the vast majority of aftershocks shift gradually to the main fault's footwall. They have focal depths mostly between 0 and 18 km, and well depict 13 NE-trending and 1 NW-trending faults with lengths of 0.6–5.7 km. Around the epicentral area, where coseismic slip is maximal (Fig. 3c profile 4), nearly all aftershocks are restricted in the main-fault's footwall. These clusters exhibit more linear alignments with shallower depths (2–8 km), revealing one long (14 km) and several short (0.7–5 km) faults. Farther north (Fig. 3c profiles 5–6), most aftershocks shift back to the main-fault's hanging wall, although some deep clusters (>18 km in depths) also appear in the footwall. In map view, these aftershocks display a more diffuse pattern without distinct alignments (Fig. 3a). A closer 3D examination, however, reveals numerous narrow seismic lineations that trend east, northeast, or north. We interpret a total of 17 faults at depths <15 km and 4 faults at >18 km, with lengths ranging from 0.5 to 5 km.

The focal mechanisms of 66 $M \geq 3.6$ events computed in this and previous studies include not only strike-slip, reverse and oblique-reverse, but some normal faulting solutions (Fig. 2a & 3c). Although many of the events occurring within or nearby clustered aftershocks have one nodal plane consistent with the aftershock-determined fault geometry, a certain number of them have no such consistent nodal plane (e.g., events 25, 59, 60 in Fig. 2a & Fig. 3c profile 4). This inconsistency may be due to large uncertainties of the focal mechanisms caused by waveform contamination, or suggest that the ruptures delineated by focal mechanisms are not imaged by aftershock alignments.

Beyond aftershock-determined subfaults, the post-mainshock InSAR data (SAR images from January 26 and February 5, 2024; see Methods) reveal two shallow oblique-reverse subfault ruptures at the immediate SW of the mainshock slip-peak area (Figs. 2b & 3c profile 3). The first one is 13 km long, 30°SE-dipping, and extending downward from the surface to the 5-km depth or deeper. It has an InSAR-inverted slip maximum of >0.6 m near the surface, which is accompanied by a >2-km-long, up to 1.0-m-thrown (SE-side-up) surface rupture zone²⁴. Such faulting kinematics agrees well with the focal mechanism (strike/dip/rake of the SE-dipping nodal plane: 40–76°/32–59°/40–116°) and the extremely shallow centroid depth of 1 km (Supplementary Fig. 7; Supplementary Data 4) for the January 29 M_w 5.8 aftershock. The second fault rupture, with a dip of 40°WNW, is spatially crossing the SE-dipping first one (Fig. 2b). This rupture has a similar lateral elongation (12 km) but a larger depth extent (>7 km), and appears to reach the surface (a slip maximum of 0.3 m) as well. Our field survey, however, identified only discontinuous, obliquely-aligned fractures (tens of

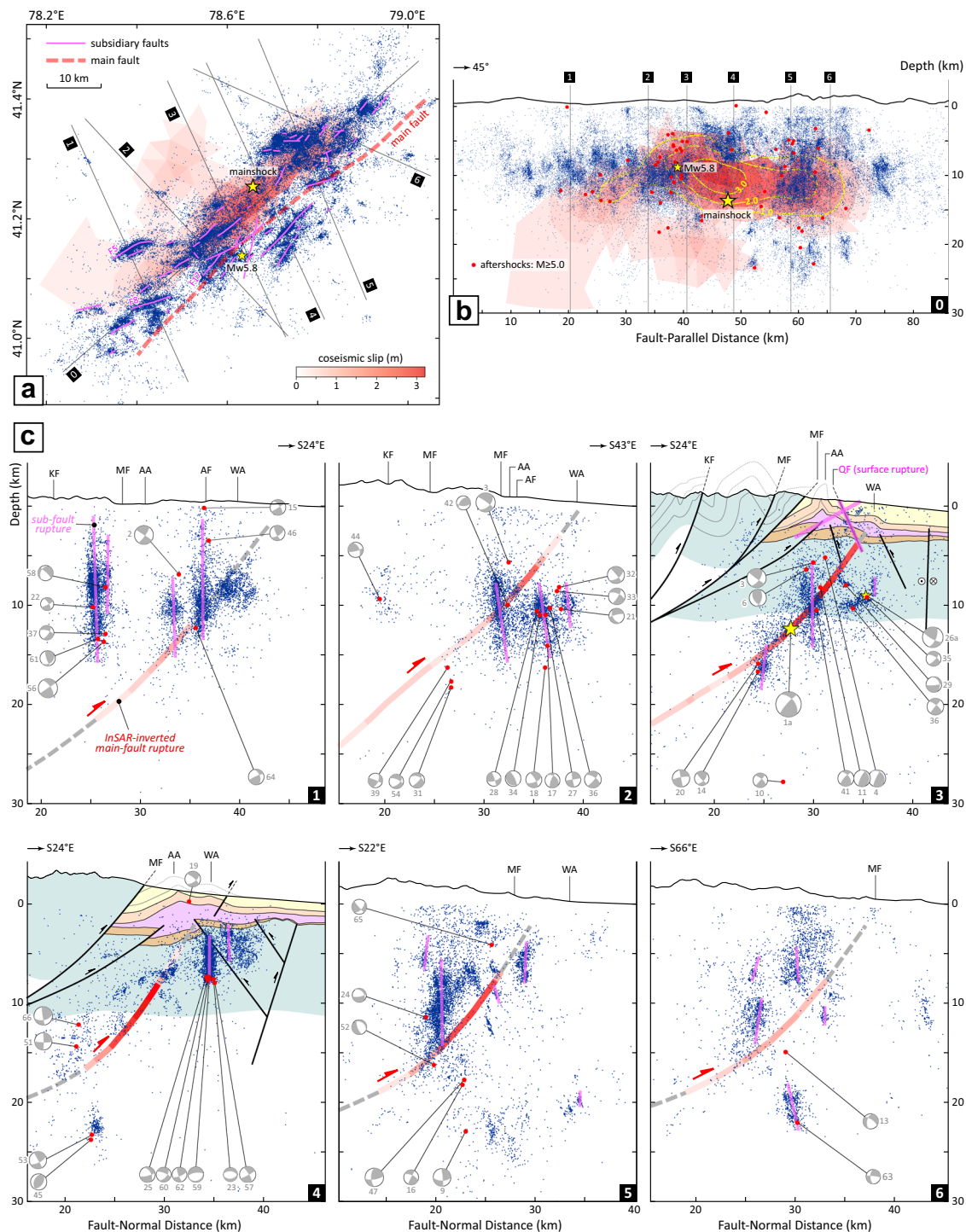


Fig. 3 | Sections of Wushi seismicity and causative main-subsidary fault geometries. **a** Map of earthquake epicenters on top of the mainshock slip distribution. The fault numbers correspond to those in Supplementary Fig. 16. Multiple section lines are marked. **b** Fault-parallel section containing the mainshock slip and all aftershocks. **c** Set of 6 cross sections (see Fig. 3a, b for locations; events within 2–4 km from the section are included) showing hypocenter distribution and fault geometries

(sub-fault ruptures shown in pink lines) at depth. Focal mechanisms (event numbers correspond to those in Fig. 2a & Supplementary Data 4) of large events ($M \geq 3.6$) are also shown. Geologic structures (faults in thick black lines) in profiles 3–4 from interpretations of petroleum seismic-reflection profiles and surface mapping data (see Methods & Supplementary Fig. 14, 15). AA Aheqi Anticline, AF Aheqi Fault, KF Kuokale Fault, MF Maidan Fault, QF Qialemati Fault, WA Wushi Anticline.

centimeters for each length) and mole tracks (<10 cm in height; Supplementary Fig. 18); the deformation zone trends northward overall, similar as the discontinuity observed on the InSAR image (Supplementary Fig. 8d, e). The InSAR-inverted fault solution does not correspond with any nearby aftershock focal mechanisms (Fig. 2a, b), leaving the exact aftershock associated with this rupture undetermined.

On seismic-reflection profiles, the Wushi Blind Fault, evidenced by faulting and folding deformation of the lower-Miocene strong seismic reflections (Fig. 1c & Fig. 3c profiles 3–4; Supplementary Fig. 14, 15), is coincident with the mainshock rupture location and mechanism, and therefore is likely the causative fault for the M_w 7.0 mainshock. The fault is restricted to below the weak middle-Miocene unit, consistent with the

observation that the mainshock rupture was terminated at 3–5 km depth as well (Figs. 2c & 3b). Based on (i) the 0.5–1.4 km difference in uplift of the lower-Miocene unit across the fault, (ii) the InSAR-inverted fault dip of 50–60° of its upper portion, and (iii) a 1:1 ratio of dip-slip and strike-slip components, the total fault slip is estimated to be 0.8–2.5 km, implying a poorly-developed, immature fault structure^{47,48}. Unlike the blind mainshock fault, the two shallow InSAR-determined subsidiary faults appear to extend from the Paleozoic bedrock to penetrate the entire Cenozoic sequence (Fig. 3c profile 3). Our seismic profiles can only image the causative fault of the M_w 5.8 aftershock (named the Qialemati Fault), which displaces the lower-Miocene unit by 200 m at depth and shows as a bedding-subparallel geometry within the Pleistocene exposures (Fig. 1c). This fault lies at the blind tip of the main fault but dips oppositely, and is likely formed in response to the main fault's upward propagation. Except for these two shallow faults, almost all of the other subsidiary faults appear to be localized within the strongly deformed Pre-Cenozoic bedrock, none of which can be identified on the seismic-reflection images (Fig. 3c).

Discussions and implications

Our integrated seismic, geodetic, and geological analyses reveal an interlacing oblique-reverse fault network ruptured during the 2024 Wushi earthquake sequence (Figs. 2c & 3). This network comprises (i) a NW-dipping, blind main fault with dips of 40–60° and (ii) numerous steeply-to-vertically dipping (69–90°) subsidiary faults of varying lengths (<1 km to >10 km) that, in cross-section, align obliquely to the main fault. The main fault is structurally immature, as evidenced by its total slip of ≤ 2.5 km. The subsidiary faults, located in both the hanging wall and footwall of the main fault, exhibit high degrees of spatial heterogeneity.

Using focal mechanisms of 66 relatively large ($M \geq 3.6$) Wushi events, stress field in the Wushi area is determined to have a nearly horizontal maximum compressive stress σ_1 (plunge 3°) orienting at 177° and a sub-vertical minimum compressive stress σ_3 (plunge 85°) (Fig. 4a; see Methods), consistent with the nearly N-S compression in the Wushi-Keping area from geodetic measurements²⁹. Under this stress field, the preferred fault structure would be thrust faults that trend perpendicular to σ_1 (approximately E-trending) and dip at around 30°, according to the Anderson's classical faulting theory⁴⁹. This idealized geometry, however, contrasts with the observed Wushi fault geometry featuring steep dips and a broad range of trends (from eastward to northward; Figs. 2c & 3). The non-Andersonian fault structure may occur in situations of domino-type rotation of initial Andersonian faults with continued bulk shortening⁵⁰, misorientation of initial Andersonian faults with stress field rotation⁵¹, or reactivation of inherited structures that are misoriented with the prevailing stress field^{50,52,53}. Given the weak Cenozoic deformation of subsalt structures in the Wushi basin and the lack of evidence for significant stress rotation in late-Cenozoic in the southern Tian Shan, the non-Andersonian Wushi faults most likely arise from the reactivation of inherited structures under the prevailing stress field. Potential inherited structures may include (i) normal or strike-slip faults formed in multiple pre-Cenozoic deformation episodes^{42,54}, (ii) high-angle reverse faults formed in Paleozoic due to strong bulk shortening^{55,56}, and (iii) bedding-parallel weaknesses due to lithologic variation and folding-related flexural slip^{57,58}, as observed along the bedding-subparallel Qialemati Fault (Fig. 1c).

In areas dominated by inherited planes of weakness, whether this plane can be reactivated or not largely depends on its orientation relative to the prevailing stress field⁵⁹. Here, we apply a slip tendency^{60,61} (T_s , equal to the ratio of shear stress acting on it to frictional resistance) analysis to evaluate the reactivation potential of the Wushi fault planes (the used frictional coefficient $\mu = 0.6$, a good approximation of the immature Wushi faults; see Methods & Supplementary Data 5). As shown in Fig. 4a, the main blind fault plane has an optimal orientation ($T_s = 0.75$), making it one of the most likely candidates for reactivation by background stress loading. This may explain why it can develop as the primary fault in the Wushi fault network. Comparatively, most subsidiary fault planes (72%) are unfavorably orientated ($T_s < 0.5$), with 20% even severely misoriented ($T_s < 0.25$). To reactivate the

plane so close to frictional lock-up, one possible mechanism is main-fault-rupture-induced Coulomb stress increase. For example, the Coulomb stress increase caused by the main fault rupture is ≥ 1.0 MPa along F15, F31–34, and F47 (Fig. 4b; see Methods & Supplementary Data 5). Because a certain number of unfavorably-oriented planes lie in the stress decrease regime (e.g., F5, F8, and F17), other reactivation mechanisms should be included, such as Coulomb stress increase due to neighboring subfault rupture, or high-pressure fluid diffusion^{60,62}, which is possible in the Wushi area due to the existence of oil/gas prospects in the Pre-Cenozoic bedrock⁴². For the remaining 28% of the subsidiary fault planes, except the M_w 5.8 aftershock plane that is optimally oriented ($T_s = 0.88$ – 0.95), all other planes correspond to be prone to, but not optimal to, reactivation ($0.5 \leq T_s < 0.75$). Their reactivation can be accomplished by background stress loading, but can also be promoted by main- or subfault-rupture-induced stress increase and/or high-pressure fluid diffusion.

The interlacing Wushi fault network formed through the reactivation of inherited planes of weakness is similar to the 2004–2007 Niigata earthquake fault network in Japan^{19,63}, but differs from fault networks of the 2000 Western Tottori earthquake in Japan¹⁶ and the 2019 Ridgecrest earthquake sequence in California⁴, where the main and subsidiary faults are orthogonally aligned and likely formed by orthogonal ductile shear in the basement⁶⁴. Fault structures typically evolve with cumulative slip toward strain localization and geometric simplification^{65,66}. For the Wushi fault network, the continuing activity of numerous subsidiary faults implies that its structural evolution has not yet reached the stage when the main fault has become totally dominant and suppressed slips along the subfaults, consistent with its immature fault geometry.

The Wushi subsidiary faults host the majority of aftershock activities: as shown in Fig. 3c, almost all aftershock clusters are concentrated along the subfaults. In comparison, aftershocks rarely occurred along the main fault. The lack of aftershock concentration along the main fault is unusual, but has been observed in several cases, e.g., the 1980 M_s 7.3 Al Asnam, Algeria⁶⁷, the 1994 M_w 6.7 Arthur's Pass, New Zealand¹⁸, and the 2016 M_w 5.2 Borrego Springs, California⁶⁸ earthquakes, suggesting that the stress on the main fault was mostly relieved by the mainshock. These aftershocks, therefore, were not triggered by continuing slip on the main fault (i.e., afterslip)^{69,70}, but by stress adjustment along subsidiary faults within the surrounding blocks. Similarly, the lack of aftershock concentration along the Qialemati Fault also suggests that the stress on this fault was mostly relieved by the M_w 5.8 event (Fig. 3c profile 3). Relative to the main fault, the Wushi aftershocks and associated subfaults show high spatial heterogeneity (Fig. 2c & 3b), which can be attributed to the heterogeneous distribution of inherited crustal structures controlling subfault reactivations. These off-main-fault aftershocks, together with InSAR-inverted mainshock slip distribution, reveal that few of them spatially overlap with the main-slip area (Fig. 2a & 3b), consistent with the hypothesis that aftershocks, no matter on- or off- the main fault, generally occur outside the main-slip area^{21–23}.

In the Wushi earthquake, another important observation is that, while the mainshock rupture did not reach the surface, the M_w 5.8 aftershock occurring 7 days later did rupture to the surface and produce a clear faulting zone (Fig. 1c). This phenomenon was first reported by Zhang et al.²⁴ based on their repeated field observations of the surface-faulting area before and after this aftershock, and can be confirmed by InSAR data analyzed here (Fig. 2b) and other studies^{25,27}. Although somewhat unexpected, this phenomenon can be easily explained by the fault structural geometry (Figs. 1c & 3c profiles 3, 4): the mainshock causative fault, with an upper tip of 3–5 km in depth, is buried beneath the weak middle-Miocene unit. In comparison, the M_w 5.8 aftershock ruptured along an antithetic subsidiary fault that cuts through the middle-Miocene unit to reach the surface, and its rupture was likely triggered by mainshock-induced stress increase (Fig. 4b). Interestingly, similar phenomenon was also observed in the 1983 Coalinga earthquake, in which the M_w 6.5 mainshock on a blind main fault did not produce surface rupture, whereas a M 5.2 aftershock 40 days later produced clear surface rupture along a high-angle back thrust^{67,71}. For reverse or oblique-reverse faults, their growth process is often accompanied by the formation of

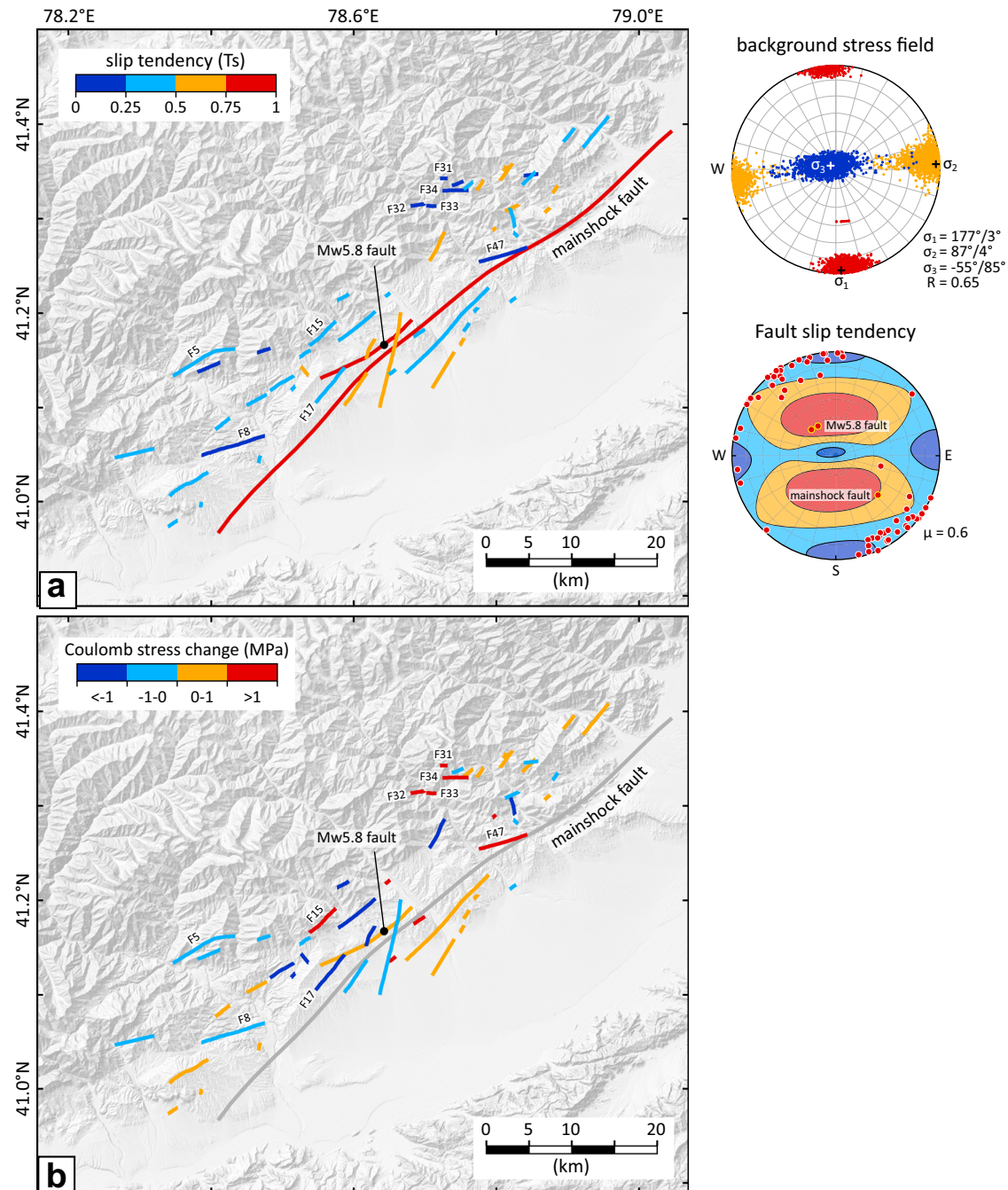


Fig. 4 | Slip tendency for Wushi main-subsidary faults and mainshock-induced Coulomb stress change. The fault numbers correspond to those in Supplementary Fig. 16. **a** Calculated fault slip tendency assuming a frictional coefficient (μ) of 0.6 (see Methods & Supplementary Data 5). The right panel shows the imposed background stress field based on focal mechanism inversions of 66 $M \geq 3.6$ Wushi events

and the slip-tendency stereo plot integrated with poles of Wushi faults. **b** Coulomb stress change on subsidiary faults (see Methods & Supplementary Data 5). The used slip (rake) angle of each subfault is estimated based on the fault orientation and the parameters of the background stress field.

subsidiary faults with varying sizes, orientations, and depth extents^{72,73}. Observations from the Wushi and Coalinga earthquakes demonstrate that, even though the main fault is totally blind, some shallowly-buried, relatively large subsidiary faults can be activated following the mainshock to produce strong, surface-rupturing aftershocks, and thereby may produce severe aftershock hazards. Such cases should be included in the forecasting of seismic hazards for further aftershocks⁷⁴.

Finally, as the largest event during the past three decades within the Tian Shan region, the 2024 Wushi earthquake offers important insights into the seismotectonic characteristics of the active Tian Shan. First, active

reverse or oblique-reverse faults within Tian Shan have typical dip angles of 35–55°, as estimated from focal mechanisms of strong earthquakes and aftershock relocations^{33,34,75}. In previous studies⁷⁶, such abnormally high-angle faults have been attributed to the reactivation of pre-existing weaknesses. Our investigation of the Wushi earthquake fault illuminates that this interpretation is at least true for some of these faults. Second, aftershock activity of the Wushi sequence has an abrupt cutoff at a depth of around 22–25 km (Fig. 3b). This depth is similar as the seismicity cutoff depth of three M_w 6.0–6.3 Jiashi earthquakes from 1998 to 2020 in the NW Tarim basin^{77–80} (Fig. 1a), indicative of similar geothermal gradients and brittle-

ductile transition depths, and therefore similar crustal strengths of the Wushi basin and the NW Tarim basin. This result is consistent with similar crustal velocity structures of these two areas from ambient noise tomography, and further supports the interpretation that the Wushi crust is the northward continuation of the Tarim craton⁸¹. The cutoff depth lies around 25 km at its deepest extent, but apparently decreases to 15–17 km toward the southwest (Fig. 3b), however, suggesting that the Wushi crust is being gradually involved into the Tian Shan mountainous crust from east to west. At last, the seismicity cutoff depth of the Wushi-NW Tarim region is much deeper than the depth of 15–17 km of the Suusamy, Ili and Bole intramountain basins^{75,82,83} (Fig. 1a), but is shallower than the depth of >30 km of the Issyk-Kul intramountain basin^{84,85}, indicating significant variation of the seismogenic thickness and the crustal strength of the Tian Shan and its adjacent regions.

Methods

Building the high-resolution aftershock catalog

We compile the 2024 Wushi earthquake catalog using continuous waveform data recorded from January 22 to March 7, 2024 by a dense seismic array (Supplementary Fig. 1). The seismic array composes of 76 three-component seismic stations, including 56 temporary stations (10 broadband and 46 short-period seismometers) installed in 2–5 days after the mainshock and 20 permanent stations (11 broadband and 9 strong-motion seismometers) of China Seismic Network and Global Seismographic Network located around the mainshock area.

The waveform data are processed by a Machine-Learning (ML) based workflow (Supplementary Fig. 2). We first perform an earthquake detection and phase picking using a recurrent neural network (RNN) deep learning model that was trained using the 2009–2020 dataset from China Seismic Network⁸⁶. We then associate the picked *P* and *S*-wave arrival times using the Rapid Earthquake Association and Location (REAL) method⁸⁷ and a 1-D layered velocity model (Supplementary Data 1) modified from the 3D Seismic Structure model⁸⁸ and the UST-Clitho2.0 model⁸⁹, during which a threshold of 8 *P* & *S* wave readings is adopted. The yielding deep-learning catalog includes 107,340 events in total. We further refine the hypocenters and calculate their associated uncertainties using the NonLinLoc program⁹⁰. The final catalog with absolute locations is made of 86,369 events (Supplementary Fig. 3a, 4a & 5a; Supplementary Data 2) with an average of root mean square (RMS) residuals of 0.12 s. We also estimate the local magnitudes (M_L) for all events. The maximum amplitudes of the horizontal component of waveforms are measured by deconvolving the instrument response from the raw waveforms and then convolving the obtained signal with the theoretical Wood-Anderson seismometer response. The estimated magnitudes range from -2.1 to 7.0 .

We determine high-precision relative locations of the events by hypoDD⁹¹ and the same 1-D velocity model as above. In this process, the differential travel time data from ML-picked arrival time and waveform cross-correlation measurements (with a coefficient of ≥ 0.7) are used. We require that each event pair has at least 8 and 5 differential travel time data from arrival time measurements and waveform cross-correlation measurements, respectively. This yields a final catalog of 59,574 events (Supplementary Fig. 3b, 4b & 5b; Supplementary Data 3). The uncertainty of relative locations is quantified using the bootstrapping method^{92,93}, in which 90% double-pair differential time data are randomly sampled for determining the relative location, and this process is repeated 100 times to obtain the standard deviation of determined relative locations. In general, the median uncertainty of relative locations is 80 m in epicenter and 91 m in focal depth. The magnitude of completeness (M_c), i.e., the magnitude above which all events are reliably detected, is estimated to be 0.5 using the maximum-curvature method with a positive 0.2 correction⁹⁴ (Supplementary Fig. 6). We also use hypoDD and cross-correlation coefficients of ≥ 0.6 to relocate these events. The obtained catalog includes 61,417 events with a median uncertainty of relative locations of 139 m in epicenter and 166 m in depth (Supplementary Fig. 3c, 4c & 5c).

GrowClust⁹⁵ is another relocation program similar to hypoDD. The algorithm uses hierarchical clustering and the L1 norm, allowing for a more robust misfit criterion than the standard least-squares approach, and can handle larger catalogs. A cross-correlation coefficient of ≥ 0.6 , a maximum station distance of 80 km, and a maximum RMS residual of 0.8 s for cluster merger are used in the algorithm, yielding a catalog of 30,732 events (Supplementary Fig. 3d, 4d & 5d). To estimate the uncertainty of relative locations, 50 bootstrap iterations of the nonparametric uncertainty estimation algorithm are performed. In general, the median uncertainties of relative location are 385 m in epicenter and 495 m in depth.

Overall, we obtain one event catalog with absolute locations and three event catalogs with relative locations. These four catalogs show similar spatiotemporal evolution and distribution patterns (Supplementary Fig. 3–5), confirming the robustness of the results. Because seismic stations are very sparse at the beginning of the earthquake sequence, and many early aftershocks cannot be well relocated by the GrowClust algorithm, we prefer to use the earthquake catalog determined using hypoDD and waveform cross-correlation coefficients of ≥ 0.7 for the subsequent analysis.

Focal mechanism and regional stress field inversions

We use the Cut-and-Paste method⁸⁶ to invert the focal mechanism of aftershocks with $M \geq 4.0$ (e.g., Supplementary Fig. 7). The waveforms are filtered using varied frequency bands according to the earthquake magnitude. For $M \geq 5.0$ aftershocks, frequency bands of 0.05–0.15 Hz for body waves and 0.05–0.1 Hz for surface waves are used. For $4.0 \leq M \leq 5.0$ earthquakes, body waves and surface waves are filtered using frequency bands of 0.05–0.17 and 0.05–0.1 Hz, respectively. The time-window lengths for the body waves and surface waves are set based on the waveform recording of the station with the maximum hypocentral distance. Our work yields a total of 18 robust solutions (Supplementary Data 4). These results are further complemented by 33 focal mechanisms of $M \geq 3.6$ aftershocks through *P*-wave polarity analyses⁹⁷, because the long-period waveform data of these events are contaminated by other aftershocks and ambient noise. Additionally, 2 focal mechanisms from the U. S. Geological Survey earthquake catalog, 2 from the Global centroid-moment tensor catalog, and 16 from the National Earthquake Data Center catalog of China are also collected. Combining these data, the regional stress field is estimated using the damped inversion method developed by Hardebeck & Michael⁹⁸ and Martínez-Garzón et al.⁹⁹. There are two nodal planes for each focal mechanism, and we randomly select one nodal plane of each focal mechanism to invert the stress field. The final stress field (Fig. 4a) is characterized by a nearly horizontal maximum compressive stress (σ_1) in the approximate N-S direction and a ratio of the principal stress differences (R) of 0.65.

InSAR data inversions

We determine fault-slip models of the Wushi mainshock and shallow aftershocks by InSAR data inversions. The collected data, including both ascending and descending Sentinel-1A and descending ALOS-2 SAR images (Supplementary Fig. 8), are processed to obtain differential interferograms using the GAMMA software^{100,101}. Topographically and atmospherically correlated radar phases are removed using the Shuttle Radar Topography Mission DEM and the ERA5 dataset, respectively. Long wavelength phase ramps, primarily caused by tropospheric/ionospheric delays and orbital errors, are removed by a flattening correction^{102,103}. The outliers with a coherence lower than 0.4 are masked before phase unwrapping to get line-of-sight (LOS) displacement using the Minimum-Cost Flow algorithm¹⁰⁴.

The obtained InSAR results not only resolve the ground deformation caused by the mainshock (Supplementary Fig. 8a–c) but by some aftershocks (Supplementary Fig. 8d, e). For the mainshock, the ground deformation is characterized by two distinct areas with opposite LOS-displacement polarities. The interferogram clearly defines the fault location and indicates a NW-dipping fault geometry based on the main uplift area that is located to the northwest of the fault (generally lying at the fault hanging wall) and the

dominantly NW-dipping fault geometries in the Wushi region (Fig. 1c). Comparatively, the deformation of aftershocks is more complicated and shows phase incoherence, generally consistent with two fault ruptures reaching the surface.

Using InSAR-measured ground deformation, the mainshock fault geometry is constructed as follows. First, the fault surface trace is interpreted following the boundary of the two distinct deformation areas (Supplementary Fig. 8). By fitting the InSAR ground deformation with an elastic dislocation model¹⁰⁵, the Bayesian inversion¹⁰⁶ of 2D fault geometry is applied along four profiles perpendicular to the fault trace (Supplementary Fig. 9), during which the fault surface trace is further refined. The fault plane along each profile is assumed to be composed by two (upper and lower) fault portions, which allows for delineating variation of the fault dip; the associated parameters for the fault plane include the horizontal location (x_0) and the vertical location (depth, y_0) of the fault lower tip, as well as the horizontal width (W_{upper} , W_{lower}) and the height (H_{upper} , H_{lower}) of each fault portion (Supplementary Fig. 9b). Finally, the 3D fault plane is constructed by interpolating controlling points of its surface trace and four profile geometries (Supplementary Fig. 10). For the two faults associated with aftershocks, the surface trace is interpreted following the trace of phase incoherence. Then, the fault plane is assumed to have a planar geometry (i.e., having a uniform dip), and a grid search is conducted to define the best-fitting fault dip combination (i.e., with the minimum residuals of the ground deformation; Supplementary Fig. 11).

Using the constructed fault geometry, the fault slip inversion is performed by fitting InSAR-observed ground deformation with the Classic Slip Inversion code¹⁰⁷ (Supplementary Fig. 12 & 13). The high-resolution InSAR data is downsampled using a quadtree partitioning approach^{108,109}. The Green's functions, which relate fault slip to the ground deformation, are computed using triangular dislocation elements in a uniform elastic half-space¹¹⁰. The relative weights of the datasets in selecting the best-fit model are determined based on the calculated covariance. Finally, the yielding fault slip is smoothed by a covariance matrix that introduces correlation between nearby parameters¹¹¹.

Interpretations of seismic-reflection profiles and relocated aftershocks

Two petroleum seismic-reflection profiles (collected in 2004 by PetroChina Company) that cut through the Wushi mainshock rupture area are used to determine the Wushi subsurface structural geometries (Supplementary Figs. 14, 15). The time-depth conversion uses a typical velocity of 5000 m/s of the Proterozoic-Paleozoic strata⁷, given that the Cenozoic strata (with a typical velocity of 3500 m/s) is very thin (≤ 1.5 s two-way travel time) and the Wushi earthquake fault is mainly developed within the Proterozoic-Paleozoic strata. This conversion will somewhat increase the depth of imaged structures, but is typically less than 1.0 km. The stratigraphic units are defined by correlating with stratal outcrops mapped by ourselves and interpretations of neighboring seismic-reflection profiles of Wang et al.⁴² using drilling data, and the geometries of major faults and folds are interpreted based on truncation or tilting of seismic reflections integrated with our surface mapping data.

The structural interpretation of Wushi relocated aftershocks is made in the SKUA-GOCAD software¹¹², which not only allows to visualize events in 3D, but also allows to select event clusters associated with different faults and fit reasonable fault planes to the clusters (Supplementary Fig. 16). We first identify event clusters that define individual fault planes based on their spatial concentration, orientation, and lateral continuity in 3D view. We then slice perpendicularly through each event cluster to construct multiple swath profiles, where nearby earthquakes are projected. On each swath profile, the fault geometry is defined by fitting a line to the projected events, and for each event cluster, a set of 2D fault lines is defined. Finally, the set of fault lines is fitted to generate a 3D fault plane. Our determining fault plane, therefore, can be considered as average of event locations. Focal mechanisms are commonly used in structural interpretation as well¹¹³. In our study,

however, this dataset has not been used because of its broad variation in slip senses and orientations (Fig. 2a).

Slip tendency analyses

For a cohesionless plane of weakness, its potential to slip or be reactivated (i.e., slip tendency T_s) depends on the ratio of shear stress (τ) acting on it to frictional resistance ($\mu \cdot \sigma_n$) of itself^{60,61}:

$$T_s = \tau / (\mu \cdot \sigma_n) = \tau / (\tan \phi \cdot \sigma_n) \quad (1)$$

where μ is the sliding friction coefficient, σ_n is the normal stress acting on the plane, and ϕ ($= \arctan \mu$) is the angle of sliding friction. Within a given stress field, the shear and normal stresses can be calculated by:

$$\sigma_n = \sigma_1 l^2 + \sigma_2 m^2 + \sigma_3 n^2 \quad (2)$$

$$\tau = [(\sigma_1 - \sigma_2)^2 l^2 m^2 + (\sigma_2 - \sigma_3)^2 m^2 n^2 + (\sigma_3 - \sigma_1)^2 l^2 n^2]^{0.5} \quad (3)$$

where σ_1 , σ_2 and σ_3 ($\sigma_1 > \sigma_2 > \sigma_3$) are the values of principal stresses, and l , m and n are the direction cosines of the plane's normal with respect to the principal stress axes, σ_1 , σ_2 and σ_3 , respectively.

In practice, stress tensor inversion can only determine the directions of the principal stress axes and the ratio of the principal stress differences $R = (\sigma_1 - \sigma_2) / (\sigma_1 - \sigma_3)$, but cannot determine the values of σ_1 , σ_2 , σ_3 , which prevents the slip tendency calculation. Considering that the frictional sliding envelope is tangential to the σ_1 - σ_3 Mohr circle (Supplementary Fig. 17), this restriction can be overcome by following conversions⁶⁰:

$$\sigma_3 = K \cdot \csc \phi - K = K(\csc \phi - 1) \quad (4)$$

$$\sigma_1 = 2K + \sigma_3 = 2K + K(\csc \phi - 1) = K(\csc \phi + 1) \quad (5)$$

$$\sigma_2 = \sigma_1 - R(\sigma_1 - \sigma_3) = K(\csc \phi + 1 - 2R) \quad (6)$$

where K is the radius of the σ_1 - σ_3 Mohr circle. By substituting Eqs. (4)–(6) in Eqs. (2) and (3), and then in Eq. (1), we can derive the slip tendency T_s that is independent of principal stress values, and depends only on R , ϕ ($= \arctan \mu$), and the principal stress orientations. For the Wushi fault planes, the $\mu = 0.6$ is used in slip tendency analysis, a good approximation of immature faults^{50,114,115}; a variation of it from 0.4 to 0.85 will not clearly change the final calculation results (Supplementary Data 5).

We also calculate the potential slip angle (rake; Supplementary Data 5) of each plane, which is coincident with the direction of shear stress on the plane, for the following estimate of mainshock-induced Coulomb stress change. The shear stress (τ) can be resolved into one component along the fault strike (τ_{strike}) and one component along the fault dip (τ_{dip})⁵²:

$$\tau_{strike} = lm(\sigma_2 - \sigma_1) / \sqrt{l^2 + m^2} \quad (7)$$

$$\tau_{dip} = n[m^2(\sigma_2 - \sigma_1) - (1 - n^2)(\sigma_3 - \sigma_1)] / \sqrt{l^2 + m^2} \quad (8)$$

Then the slip angle θ , equal to the angle between the direction of the shear stress and the fault plane strike, can be calculated by:

$$\tan \theta = \frac{\tau_{dip}}{\tau_{strike}} = \frac{n}{lm} \left[m^2 - (1 - n^2) \frac{\sigma_3 - \sigma_1}{\sigma_2 - \sigma_1} \right] = \frac{n}{lm} \left[m^2 - \frac{1 - n^2}{R} \right] \quad (9)$$

Mainshock-induced Coulomb stress change

Coulomb failure stress change¹ (ΔCFS) induced by the mainshock rupture is calculated by (Supplementary Data 5):

$$\Delta CFS = \Delta \tau - \mu' \Delta \sigma_n \quad (10)$$

where $\Delta \tau$ is the shear stress change resolved in the slip direction of the target fault, $\Delta \sigma_n$ is the normal stress change on the fault, and μ' is the apparent friction coefficient (0.6 is assumed) of the fault. The strike and dip of the target fault are from aftershock interpretations and InSAR data inversions (Supplementary Data 5). The rake of the target fault cannot be determined based on aftershock focal mechanisms because of their broad variation; here, it is represented by the potential slip angle (Eq. (9)) of the fault plane under the prevailing stress field. Fault slip potential is enhanced if $\Delta CFS > 0$ or is inhibited if $\Delta CFS < 0$.

Data availability

The Supplementary Data 1–5 are deposited at <https://doi.org/10.5281/zenodo.17160179>¹⁶, and the Supplementary Figs. 1–18 are provided in Supplementary Information. The teleseismic inversion of the 2024 Wushi mainshock, and the epicenters and focal mechanisms of major earthquakes within the Tian Shan and adjacent regions are available from the U. S. Geological Survey (<http://earthquake.usgs.gov/earthquakes/>). The seismic data from the Global Seismographic Network (GSN) are downloaded from the Earthscope Inc. Data Management Center (<https://www.iris.edu/hq/>). The raw SAR data are downloaded from Copernicus (<https://scihub.copernicus.eu/>).

Code availability

The RNN algorithm⁸⁶ is available at <https://gitee.com/cangyeone/csnbench>, the REAL algorithm⁸⁷ is available at <https://github.com/Dal-mzhang/REAL.git>, the NonLinLoc algorithm⁹⁰ is available at <http://alomax.free.fr/nllloc>, the hypoDD algorithm⁹¹ is available at http://www.ldeo.columbia.edu/~felixw/HYPDD/HYPDD_1.3.tar.gz, the GrowClust algorithm⁹⁵ is available at <https://github.com/dtrugman/GrowClust.git>, the Cut-and-Paste algorithm⁹⁶ is available at <https://www.eas.slu.edu/People/LZhu/home.html>, the Flower2D algorithm¹⁰⁶ is available at <https://github.com/simondaout/Flower2d>, the Classic Slip Inversion code¹⁰⁷ is available at <https://github.com/jolivet/csi>.

Received: 16 April 2025; Accepted: 30 September 2025;

Published online: 17 November 2025

References

- Scholz, C. The mechanics of earthquakes and faulting. Cambridge University Press (2019).
- Haeussler, P. et al. Surface rupture and slip distribution of the Denali and Totschunda faults in the 3 November 2002 M 7.9 earthquake Alaska. *Bull. Seismol. Soc. Am.* **94**, S23–S52 (2004).
- Peng, Z. et al. Mainshock rupture properties, aftershock activities and remotely triggered seismicity associated with the 2025 M_w7.7 Sagaing fault earthquake in Myanmar. *Earthq. Res. Adv.* <https://doi.org/10.31223/x5bq7d> (2025).
- Ross, Z. et al. Hierarchical interlocked orthogonal faulting in the 2019 Ridgecrest earthquake sequence. *Science* **366**, 346–351 (2019).
- Xu, S. & Ben-Zion, Y. Numerical and theoretical analyses of in-plane dynamic rupture on a frictional interface and off-fault yielding patterns at different scales. *Geophys. J. Int.* **193**, 304–320 (2013).
- Eberhart, P. & Reasenberg, P. Complex faulting structure inferred from local seismic observations of M_≥1.0 aftershocks, May 2–June 30, 1983. The Coalinga, California, Earthquake of May 2, 1983, edited by Rymer, M. & Ellsworth, W., U.S. Geol. Surv. Prof. Pap., 1487, 171–192 (1990).
- Lü, L. et al. Active structural geometries and their correlation with moderate (M 5.5–7.0) earthquakes in the Jiashi–Keping region, Tian Shan southwestern front. *Tectonics* **40**, e2021TC006760 (2021).
- Ren, C. et al. Supershear triggering and cascading fault ruptures of the 2023 Kahramanmaraş, Türkiye, earthquake doublet. *Science* **383**, 305–311 (2024).
- Stein, R. & Bird, P. Why do great continental transform earthquakes nucleate on branch faults? *Seismol. Res. Lett.* **95**, 3406–3415 (2024).
- Chuang, L. et al. Foreshocks of the 2010 Mw 6.7 Yushu, China Earthquake occurred near an extensional step-over. *JGR: Solid Earth*, 128, e2022JB025176 (2023).
- Hudnut, K., Seeber, L. & Pacheco, J. Cross-fault triggering in the 24 November 1987 Superstition Hill earthquake sequence, southern California. *Geophys. Res. Lett.* **16**, 199–202 (1989).
- Neves, M. et al. Complex rupture dynamics of the extremely shallow August 2020 M5.1 Sparta, North Carolina earthquake. *Commun. Earth Environ.* **5**, 163 (2024).
- Peng, Z. et al. The Evolution Process between the Earthquake Swarm Beneath the Noto Peninsula, Central Japan and the 2024 M 7.6 Noto Hanto Earthquake Sequence. *Earthq. Res. Adv.* <https://doi.org/10.1016/j.eqrea.2024.100332> (2024).
- Valoroso, L. et al. Radiography of a normal fault system by 64,000 high-precision earthquake locations: The 2009 L'Aquila (central Italy) case study. *J. Geophys. Res.* **118**, 1156–1176 (2013).
- Yoshida, K. et al. (2024). Role of a hidden fault in the early process of the 2024 M_w 7.5 Noto Peninsula earthquake. *Geophys. Res. Lett.* **51**, e2024GL110993 (2024).
- Fukuyama, E., Ellsworth, W. L., Waldhauser, F. & Kubo, A. Detailed fault structure of the 2000 Western Tottori, Japan, earthquake sequence. *Bull. Seismol. Soc. Am.* **93**, 1468–1478 (2003).
- Hauksson, E., Jones, L. & Hutton, K. The 1999 M_w 7.1 Hector Mine, California, Earthquake sequence: Complex conjugate strike-slip faulting. *Bull. Seismol. Soc. Am.* **92**, 1154–1170 (2002).
- Robinson, R. & McGinty, P. The enigma of the Arthur's Pass, New Zealand, earthquake 2. The aftershock distribution and its relation to regional and induced stress fields. *J. Geophys. Res.* **105**, 16139–16150 (2000).
- Sibson, R. An episode of fault-valve behaviour during compressional inversion? — The 2004 M_j 6.8 Mid-Niigata Prefecture, Japan, earthquake sequence. *Earth Planet. Sci. Lett.* **257**, 188–199 (2007).
- Waldhauser, F., Michele, M., Chiaraluce, L., Di Stefano, R. & Schaff, D. Fault planes, fault zone structure and detachment fragmentation resolved with high-precision aftershock locations of the 2016–2017 central Italy sequence. *Geophys. Res. Lett.* **48**, e2021GL02918 (2021).
- Das, S. & Henry, C. Spatial relation between main earthquake slip and its aftershock distribution. *Rev. Geophys.* **41**, 1013 (2003).
- Mendoza, C. & Hartzell, S. Aftershock patterns and main shock faulting. *Bull. Seismol. Soc. Am.* **78**, 1438–1449 (1988).
- Wetzler, N., Lay, T., Brodsky, E. & Kanamori, H. Systematic deficiency of aftershocks in areas of high coseismic slip for large subduction zone earthquakes. *Sci. Adv.* **4**, eaao3225 (2018).
- Zhang, B. et al. Geological disasters and surface ruptures of January 23, 2024 M_s 7.1 Wushi earthquake, Xinjiang, China. *Seismol. Geol.* **46**, 220–234 (2024).
- Qiu, J., Sun, J. & Ji, L. The 2024 Mw 7.1 Wushi Earthquake: A thrust and strike-slip event unveiling the seismic mechanisms of the South Tian Shan's thick-skin tectonics. *Remote Sens.* **16**, 2937 (2024).
- Zheng, R., Zou, R., Dong, R., Fang, Z. & Wang, Q. The 2024 M_w 7.0 Wushi Earthquake in Southern Tianshan convergent zone: Finite-fault model for the coseismic rupture and aftershock. *Seismol. Res. Lett.* **96**, 816–827 (2024).
- Thompson Jobe, J. et al. Surface rupture from an aftershock: Remote observations from the January 2024 Wushi (Aykol), China,

- Earthquakes. *Seismol. Res. Lett.* <https://doi.org/10.1785/0220240347> (2025).
28. Molnar, P. & Tapponnier, P. Cenozoic tectonics of Asia: Effects of a continental collision. *Science* **189**, 419–426 (1975).
 29. Wang, M. & Shen, Z. Present-day crustal deformation of continental China derived from GPS and its tectonic implications. *J. Geophys. Res.: Solid Earth* **125**, e2019JB018774 (2020).
 30. Wu, C. et al. Slip partitioning and crustal deformation patterns in the Tianshan orogenic belt derived from GPS measurements and their tectonic implications. *Earth-Sci. Rev.* **238**, 104362 (2023).
 31. Yang, S., Li, J. & Wang, Q. The deformation pattern and fault rate in the Tianshan Mountains inferred from GPS observations. *Sci. China Ser. D: Earth Sci.* **51**, 1064–1080 (2008).
 32. Zubovich, A. et al. GPS velocity field of the Tien Shan and surrounding regions. *Tectonics* **29**, 1–23 (2010).
 33. Ghose, S., Hamburger, M. & Ammon, C. Source parameters of moderate-size earthquakes in the Tien Shan, Central Asia from regional moment tensor inversion. *Geophys. Res. Lett.* **25**, 3181–3184 (1998).
 34. Nelson, M., McCaffery, R. & Molnar, P. Source parameters for 11 earthquakes in the Tien Shan, central Asia, determined by P and SH waveform inversion. *J. Geophys. Res.* **92**, 12629–12648 (1987).
 35. Thompson, S. et al. Late Quaternary slip rates across the central Tien Shan, Kyrgyzstan, central Asia. *J. Geophys. Res.* **107**, 2203 (2002).
 36. Avouac, J., Tapponier, P., Bai, M., You, H. & Wang, G. Active thrusting and folding along the northern Tien Shan and late Cenozoic rotation of the Tarim relative to Dzungaria and Kazakhstan. *J. Geophys. Res.* **98**, 6755–6804 (1993).
 37. Heermance, R., Chen, J., Burbank, D. & Miao, J. Temporal constraints and pulsed Late Cenozoic deformation during the structural disruption of the active Kashi foreland, northwest China. *Tectonics* **27**, TC6012 (2008).
 38. Hubert-Ferrari, A., Suppe, J., Van Der Woerd, J., Wang, X. & Lu, H. Irregular earthquake cycle along the southern Tianshan front, Aksu area, China. *J. Geophys. Res.* **110**, B06402 (2005).
 39. Kulikova, G. Source parameters of the major historical earthquakes in the Tien-Shan region from the late 19th to the early 20th century. PhD thesis, University of Potsdam. (2016).
 40. Molnar, P. & Ghose, S. 2000. Seismic moments of major earthquakes and the rate of shortening across the Tien Shan. *Geophys. Res. Lett.* **27**, 2377–2380 (2000).
 41. Jia, C., Zhang, S. & Wu, S. Stratigraphy of the Tarim Basin and Adjacent Areas (p. 1063). Science Press. (2004).
 42. Wang, Q., Yang, W., Zhou, H. & Miao, W. The stratigraphic system and structural characteristics of the western subsag of Wushi Sag, NW Tarim Basin. *Chin. J. Geol.* **59**, 271–287 (2024).
 43. Wu, C. et al. Oblique thrust of the Maidan fault and late Quaternary tectonic deformation in the southwestern Tian Shan, northwestern China. *Tectonics* **38**, 2625–2645 (2019).
 44. Wang, X. et al. Distribution of interseismic coupling along the Maidan Fault in Tianshan before the 2024 Mw 7.0 Wushi earthquake. *Geophys. Res. Lett.* **51**, e2024GL111472 (2024).
 45. Allen, M., Windley, B. & Chi, Z. Late Cenozoic tectonics of the Kepingtage thrust zone: Interactions of the Tien Shan and Tarim Basin, northwest China. *Tectonics* **18**, 639–654 (1999).
 46. Zheng, M. et al. Structural pattern and its control on hydrocarbon accumulations in Wushi Sag, Kuche Depression, Tarim Basin. *Pet. Explor. Dev.* **35**, 444–451 (2008).
 47. Manighetti, I., Mercier, A. & De Barros, L. Fault trace corrugation and segmentation as a measure of fault structural maturity. *Geophys. Res. Lett.* **48**, e2021GL095372 (2021).
 48. Perrin, C., Manighetti, I., Ampuero, J., Cappa, F. & Gaudemer, Y. Location of largest earthquake slip and fast rupture controlled by along-strike change in fault structural maturity due to fault growth. *J. Geophys. Res.: Solid Earth* **121**, 3666–3685 (2016).
 49. Anderson, E. The Dynamics of Faulting and Dyke Formation with Application to Britain (p. 206). 2nd ed. Edinburgh: Oliver & Boyd (1951).
 50. Sibson, R. Rupture nucleation on unfavorably oriented faults. *Bull. Seismol. Soc. Am.* **80**, 1580–1604 (1990).
 51. Williams, A. Oblique-slip faults and rotated stress systems. *Geol. Mag.* **95**, 207–218 (1958).
 52. Bott, M. The mechanisms of oblique slip faulting. *Geol. Mag.* **96**, 109–117 (1959).
 53. Nevitt, J., Brooks, B., Hardebeck, J. & Aagaard, B. 2019 M7.1 Ridgecrest earthquake slip distribution controlled by fault geometry inherited from Independence dike swarm. *Nat. Commun.* **14**, 1546 (2023).
 54. Wu, H. et al. Ediacaran magmatism and rifting along the northern margin of the Tarim craton: Implications for the late Neoproterozoic Rodinia configuration and breakup. *Geol. Soc. Am. Bull.* **135**, 367–388 (2023).
 55. Windley, B., Allen, M., Zhang, C., Zhao, Z. & Wang, G. Paleozoic accretion and Cenozoic redeformation of the Chinese Tien Shan Range, Central Asia. *Geology* **18**, 128–131 (1990).
 56. Xiao, W., Windley, B., Allen, M. & Han, C. Paleozoic multiple accretionary and collisional tectonics of the Chinese Tianshan orogenic collage. *Gondwana Res.* **23**, 1316–1341 (2013).
 57. Li, T., Chen, J., Thompson Jobe, J. & Burbank, D. Active flexural-slip faulting: Controls exerted by stratigraphy, geometry, and fold kinematics. *J. Geophys. Res.: Solid Earth* **122**, 8538–8565 (2017).
 58. Philip, H. & Meghraoui, M. Structural analysis and interpretation of the surface deformations of the El Asnam earthquake of October 10, 1980. *Tectonics* **2**, 17–49 (1983).
 59. Sibson, R. A note on fault reactivation. *J. Struct. Geol.* **7**, 751–754 (1985).
 60. Lisle, R. & Srivastava, D. Test of the frictional reactivation theory for faults and validity of fault-slip analysis. *Geology* **32**, 569–572 (2004).
 61. Morris, A., Ferrill, D. & Henderson, D. Slip-tendency analysis and fault reactivation. *Geology* **24**, 275–278 (1996).
 62. Miller, S. Aftershocks are fluid-driven and decay rates controlled by permeability dynamics. *Nat. Commun.* **11**, 5787 (2020).
 63. Kato, A. et al. Reactivation of ancient rift systems triggers devastating intraplate earthquakes. *Geophys. Res. Lett.* **36**, L05301 (2009).
 64. Thatcher, W. & Hill, D. Fault orientations in extensional and conjugate strike-slip environments and their implications. *Geology* **19**, 1116–1120 (1991).
 65. Ben-Zion, Y. & Sammis, C. Characterization of fault zones. *Pure Appl. Geophys.* **160**, 677–715 (2003).
 66. Perrin, C., Waldhauser, F. & Scholz, C. The shear deformation zone and the smoothing of faults with displacement. *J. Geophys. Res.: Solid Earth* **126**, e2020JB020447 (2021).
 67. Ouyed, M., Yielding, G., Hatzfeld, D. & King, G. An aftershock study of the El Asnam (Algeria) earthquake of 1980 October 10. *Geophys. J. R. Astron. Soc.* **73**, 605–639 (1983).
 68. Ross, Z., Hauksson, E. & Ben-Zion, Y. Abundant off-fault seismicity and orthogonal structures in the San Jacinto fault zone. *Sci. Adv.* **3**, e1601946 (2017).
 69. Peng, Z. & Zhao, P. Migration of early aftershocks following the 2004 Parkfield earthquake. *Nat. Geosci.* **2**, 877–881 (2009).
 70. Perfettini, H., Frank, W. B., Marsan, D. & Bouchon, M. A model of aftershock migration driven by afterslip. *Geophys. Res. Lett.* **45**, 2283–2293 (2018).
 71. Rymer, M., Kendrick, K., Lienkaemper, J. & Clark, M. The Nuñez fault and its surface rupture during the Coalinga earthquake sequence, in The Coalinga, California, Earthquake of May 2, 1983, edited by Rymer, M. & Ellsworth, W. *U. S. Geol. Surv. Prof. Pap.* **1487**, 299–318 (1990).

72. Li, T. et al. Active bending-moment faulting: Geomorphic expression, controlling conditions, accommodation of fold deformation. *Tectonics* **37**, 2278–2306 (2018).
73. Tavani, S. et al. A review of deformation pattern templates in foreland basin systems and fold-and-thrust belts: Implications for the state of stress in the frontal regions of thrust wedges. *Earth-Sci. Rev.* **141**, 82–104 (2015).
74. Hardebeck, J. et al. Aftershock forecasting. *Annu. Rev. Earth Planet. Sci.* **52**, 61–84 (2024). 1146/annurev-earth-040522-102129.
75. Mellors, R. et al. The $M_s = 7.3$ 1992 Suusamy, Kyrgyzstan, earthquake: 1. Constraints on fault geometry and source parameters based on aftershocks and body-wave modeling. *Bull. Seismol. Soc. Am.* **87**, 11–22 (1997).
76. Sibson, R. & Xie, G. Dip range for intracontinental reverse fault ruptures: Truth not stranger than friction?. *Bull. Seismol. Soc. Am.* **88**, 1014–1022 (1998).
77. Guo, B., Liu, Q., Chen, J. & Li, S. Relocation of earthquake sequences using joint hypocenter determination method: Portable seismic array study in Jiashi region, Xinjiang (in Chinese). *Seismol. Geol.* **24**, 199–207 (2002).
78. Huang, Y., Yang, J. & Zhang, T. Relocation of the Bachu-Jiashi, Xinjiang earthquake sequence in 2003 using the double-difference location algorithm. *Chin. J. Geophys.* **49**, 162–169 (2006).
79. Zhang, Y. et al. Strain accommodation and seismic hazards of the Kalpin fold-and-thrust belt, southwestern Tian Shan foreland, China: Insights from the 2020 M_w 6.0 Kalpin earthquake. *J. Geophys. Res.: Solid Earth* **128**, e2022JB025328 (2023).
80. Zhao, C., Chen, Z., Zheng, S. & Zhang, Z. Moment inversion of moderate earthquakes and the locally perturbed stress field in the Jiashi source region. *Chin. J. Geophys.* **51**, 782–792 (2008).
81. Sun, W. et al. Forced Cenozoic continental subduction of Tarim craton-like lithosphere below the Tianshan revealed by ambient noise tomography. *Geology* **50**, 1393–1397 (2022).
82. Fang, L., Wu, J., Wang, C., Wang, W. & Yang, T. Relocation of the 2012 M_s 6.6 Xinjiang Xinyuan earthquake sequence. *Sci. China: Earth Sci.* **56**, 1–5 (2014).
83. Hu, Z., Li, T., Thompson Jobe, J. & Yang, X. Lower portion rupture of a thrust fault during the 2017 M_w 6.3 Jinghe earthquake: Implications to seismic hazards in the Tian Shan region. *Seismol. Res. Lett.* **92**, 2134–2144 (2021).
84. Roecker, S. et al. Three-dimensional elastic wave velocity structure of the western and central Tien Shan. *J. Geophys. Res.* **98**, 15779–15795 (1993).
85. Xu, Y., Roecker, S., Wei, R., Zhang, W. & Wei, B. Analysis of seismic activity in the crust from earthquake relocation in the central Tien Shan. *Bull. Seismol. Soc. Am.* **96**, 737–744 (2006).
86. Yu, Z., Wang, W. & Chen, Y. Benchmark on the accuracy and efficiency of several neural network based phase pickers using datasets from China Seismic Network. *Earthq. Sci.* **36**, 113–131 (2022).
87. Zhang, M., Ellsworth, W. & Beroza, G. Rapid earthquake association and location. *Seismol. Res. Lett.* **90**, 2276–2284 (2019).
88. Xiao, X. et al. Shallow seismic structure beneath the continental China revealed by P-wave polarization, Rayleigh wave ellipticity and receiver function. *Geophys. J. Int.* **225**, 998–1019 (2021).
89. Han, S., Zhang, H., Xin, H., Shen, W. & Yao, H. USTClitho2.0: Updated unified seismic tomography models for Continental China lithosphere from joint inversion of body-wave arrival times and surface-wave dispersion data. *Seismol. Soc. Am.* **93**, 201–215 (2022).
90. Lomax, A., Michelini, A. & Curtis, A. Earthquake location, direct, global-search methods. In Meyers, R. (Ed.), *Encyclopedia of complexity and systems science* (pp. 1–33). Springer. https://doi.org/10.1007/978-3-642-27737-5_150-2 (2014).
91. Waldhauser, F. & Ellsworth, W. A. double-difference earthquake location algorithm: Method and application to the northern Hayward fault, California. *Bull. Seismol. Soc. Am.* **90**, 1353–1368 (2000).
92. Efron, B. & Gong, G. A leisurely look at the bootstrap, the jackknife, and cross-validation. *Am. Stat.* **37**, 36–48 (1983).
93. Efron, B. & Tibshirani, R. Statistical data analysis in the computer age. *Science* **253**, 390–395 (1991).
94. Woessner, J. & Wiemer, S. Assessing the quality of earthquake catalogues: Estimating the magnitude of completeness and its uncertainty. *Bull. Seismol. Soc. Am.* **95**, 684–698 (2005).
95. Trugman, D. & Shearer, P. GrowClust: A hierarchical clustering algorithm for relative earthquake relocation, with application to the Spanish Springs and Sheldon, Nevada, earthquake sequences. *Seismol. Res. Lett.* **88**, 379–391 (2017).
96. Zhu, L. & Helmberger, D. Advancement in source estimation techniques using broadband regional seismograms. *Bull. Seismol. Soc. Am.* **86**, 1634–1641 (1996).
97. Wan, Y., Wu, Y., Sheng, S., Shen, Z. & Wan, D. Preliminary result of Taiwan 3-D stress field from P wave polarity data. *Chin. J. Geophys.* (in Chinese), 54, 2809–2818
98. Hardebeck, J. & Michael, A. Damped regional-scale stress inversions: Methodology and examples for Southern California and the Coalinga aftershock sequence. *J. Geophys. Res.: Solid Earth* **111**, 1–11 (2006).
99. Martínez-Garzón, P., Kwiath, G., Ickrath, M. & Bohnhoff, M. MSATSI: A MATLAB package for stress inversion combining solid classic methodology, a new simplified user-handling, and a visualization tool. *Seismol. Res. Lett.* **85**, 896–904 (2014).
100. Wegmüller, U. et al. Sentinel-1 support in the GAMMA software. *Procedia Comput. Sci.* **100**, 1305–1312 (2016).
101. Werner, C., Wegmüller, U., Strozzi, T. & Wiesmann, A. GAMMA SAR and interferometric processing software, in Proceedings of the ERS-Envisat Symposium, Gothenburg (2001).
102. Fattahi, H., Simons, M. & Agram, P. InSAR time-series estimation of the ionospheric phase delay: An extension of the split range-spectrum technique. *IEEE Trans. Geosci. Remote Sens.* **55**, 5984–5996 (2017).
103. Gomba, G., González, F. & De Zan, F. Ionospheric phase screen compensation for the Sentinel-1 TOPS and ALOS-2 ScanSAR modes. *IEEE Trans. Geosci. Remote Sens.* **55**, 223–235 (2016).
104. Werner, C., Wegmüller, U., Strozzi, T. & Wiesmann, A. Processing strategies for phase unwrapping for InSAR applications, in Proceedings of the European Conference on synthetic aperture radar (2002).
105. Segall, P. Earthquake and volcano deformation. Princeton, NJ: Princeton University Press. <https://doi.org/10.1002/0471743984.vse7429> (2010).
106. Daout, S. et al. Constraining the kinematics of metropolitan Los Angeles faults with a slip-partitioning model. *Geophys. Res. Lett.* **43**, 11192–11201 (2016).
107. Jolivet, R. et al. Daily to centennial behavior of aseismic slip along the central section of the North Anatolian Fault. *J. Geophys. Res.: Solid Earth* **128**, e2022JB026018 (2023).
108. Lohman, R. & Simons, M. Some thoughts on the use of InSAR data to constrain models of surface deformation: Noise structure and data downsampling. *Geochim. Geophys. Geosyst.* **6**, Q01007 (2005).
109. Sudhaus, H. & Sigurjón, J. Improved source modeling through combined use of InSAR and GPS under consideration of correlated data errors: application to the June 2000 Kleifarvatn earthquake, Iceland. *Geophys. J. Int.* **176**, 389–404 (2009).
110. Meade, B. Algorithms for the calculation of exact displacements, strains, and stresses for triangular dislocation elements in a uniform elastic half space. *Comput. Geosci.* **33**, 1064–1075 (2007).

111. Radiguet, M. et al. Spatial and temporal evolution of a long term slow slip event: the 2006 Guerrero Slow Slip Event. *Geophys. J. Int.* **184**, 816–828 (2011).
112. Carena, S. & Suppe, J. Three-dimensional imaging of active structures using earthquake aftershocks: The Northridge thrust, California. *J. Struct. Geol.* **24**, 887–904 (2002).
113. Plesch, A., Shaw, J., Ross, Z. & Hauksson, E. Detailed 3D Fault Representations for the 2019 Ridgecrest, California, Earthquake Sequence. *Bull. Seismol. Soc. Am.* **110**, 1818–1831 (2020).
114. Collettini, C. & Trippetta, F. A slip tendency analysis to test mechanical and structural control on aftershock rupture planes. *Earth Planet. Sci. Lett.* **255**, 402–413 (2007).
115. Yukutake, Y., Takeda, T. & Yoshida, A. The applicability of frictional reactivation theory to active faults in Japan based on slip tendency analysis. *Earth Planet. Sci. Lett.* **411**, 188–198 (2015).
116. Yin, X. et al. Interlacing ruptures of the 2024 Wushi earthquake (Chinese Tian Shan) controlled by structural inheritance [Data set]. Zenodo, <https://doi.org/10.5281/zenodo.17160179> (2025).

Acknowledgements

We thank Wei S. J., Zhou Y. S., Wang L. F., Chen J. Y., Cheng Y. F., He X. H., Zhang Z. Q., Chen Y. K., Zhang Z., Deng F. Y. & Zhai Q. S. for fruitful discussions. This study was supported by the National Key Research and Development Program of China (Grant Number 2022YFC3003700), by the Institute of Geology, China Earthquake Administration (Grant Number IGCEA2405), and by the PetroChina Major Science and Technology Project (Grant number 2023ZZ0202). This manuscript was improved significantly by thoughtful comments and suggestions from A. Savvaidis and two anonymous reviewers.

Author contributions

X.Z.Y., T.L., Y.F.Z., Z.G.P. & L.D.Z. conceived the study and guided the scientific discussions. X.Z.Y. & Z.G.P. performed waveform-based aftershock acquisitions and relocations. T.L., Z.X.C., L.Q., X.S., R.Q.L., W.X.Y. & J.C. conducted geologic field mapping, aftershock-based structural interpretations, and seismic-reflection profiles interpretations. Y.F.Z. analyzed InSAR data and calculated Coulomb stress change due to the mainshock. T.L. & L.Q. performed slip tendency analyses. H.C. calculated focal mechanisms and the regional stress field. B.G. & J-H.C. installed the temporary seismic network. T.L. wrote most of the manuscript. All authors contributed to the writing of the paper.

Competing interests

The authors declare no competing interests. Luca Dal Zilio is an Editorial Board Member for Communications Earth & Environment, but was not involved in the editorial review of, nor the decision to publish this article.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s43247-025-02855-4>.

Correspondence and requests for materials should be addressed to Tao Li or Yingfeng Zhang.

Peer review information *Communications Earth and Environment* thanks Alexandros Savvaidis and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. Primary Handling Editor: Joe Aslin. [A peer review file is available.]

Reprints and permissions information is available at <http://www.nature.com/reprints>

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2025